

Traffic Sign Recognition and Classification Using YOLOv2, Faster RCNN and SSD

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Abstract— In the walk of Advanced Driving Assistance Systems (ADAS), Intelligent Driving and Traffic safety, Object detection plays a crucial role in the upcoming genesis of self-governing vehicles. Traditional computer vision and machine learning advances for object detection confront challenges against the difficult image backgrounds and environment conditions like sunlight effects, barricades and occlusions. In this paper, Single Shot Detector (SSD), Faster Region Convolutional Neural Network (Faster RCNN) and You Only Look Once (YOLOv2) deep learning architectures are compared by applying distinct pretrained Convolutional Neural Network (CNN) models. Experiments have been organized in a wide range to attain distinct models of Faster RCNN, SSD and YOLOv2 through appropriate modification in algorithms and parameters tuning. In this work, SSD, Faster RCNN and YOLOv2 are trained for 5 different object classes of traffic signs and their outcomes are evaluated. Traditional Evaluation parameters: mAp(mean Average precision-Precision, Recall and IoU) and FPS(Frames per second) are run-down to analyze the accuracy and speed of the algorithms. On analyzing, the accuracy of YOLOv2 outperforms Faster RCNN and SSD by 3.5% and 21% respectively. Also, YOLOv2 learned 3 times speedy than Faster RCNN with increased accuracy.

Index Terms— Traffic Sign detection, YOLOv2, Faster R-CNN, Pre-trained CNN, SSD, TensorFlow.

I. INTRODUCTION

DUE to advancements of civilization, traffic is expanding enormously and henceforth, the rate of Accidents is increasing vastly. A study conducted by Global Health Observatory of World Health Organization 2018 states that, approximately 1.35 million inhabitants died due to road accidents last year and the leading cause highlighted was Traffic violations [1]. The innovation of Advanced Driver Assistance Systems (ADAS) in the field of Computer Vision comes up to be a great measure to unfold the issue. Intelligent recognition and classification of Traffic Signs is one of the core pieces of the ADAS technology.

The main hardships to be beaten to attain our objective are different camera viewpoints, image scales, colour abnormality, barricades and weather influences like lightening, sunlight impressions and wind effects. Conventional object detection methods which include Hand crafted feature extraction methods

like Histogram of Oriented Graphs (HOG) [2], Speeded-Up Robust Features (SURF) and Scale Invariant Feature Transform (SIFT) achieved great performance but failed to overcome the above-mentioned difficulties [3].

In previous years, the evolution of Convolutional Neural Networks (CNN) in the domain of object-detection has performed profoundly strong/well. The modern algorithms used CNN to extract features and Support Vector Machine (SVM) to classify the objects. Observing positive results from CNN, numerous researchers started working in this field to attain improved accuracy and faster algorithms. The latest approaches discovered in object detection include Region CNN(RCNN) [4][5], Fast RCNN [6] and Faster RCNN [7]. These advances bring prognosis on the order of hundreds of milliseconds to seconds. In the search of faster algorithm, YOLO [8] (You Only Look Once) which is six times faster than Faster RCNN, is considered in this paper.

In this work, we have used YOLOv2 [9] to detect and recognize traffic signs. We have used multiple hyperparameters to detect traffic sign bounding boxes and their classification. Also, we have compared the results with SSD [10] and Faster RCNN architectures. This paper is systematized as follows: Section II illustrates the related work done in the object detection domain. Section III describes SSD, Faster RCNN and YOLOv2 architectures. Section IV demonstrates the results obtained from these architectures and Section V presents the conclusion.

II. RELATED WORK

Object detection is thriving in the field of computer vision and is advancing swiftly. Object detection approaches comprises of background modelling methods, template matching methods, hand-crafted feature methods and CNN based methods. Background modelling-based methods performed strong for running objects, but were unsuccessful with the immobile objects [11]. The method based on template matching exploits the information like image texture, contour and grayscale for the task of object detection. The dominance of Template matching methods is their independency with feature extraction as they perform with raw image [12]. Down the years, Hand crafted features like Histogram of Oriented Graphs (HOG), SURF, Hough Transform, KAZE and TILDE were used for object detection [13][14]. The achievement of

object detection task relies upon the quality of extracted features. All valuable Feature extraction algorithms are reviewed and abstracted in table 1[15][16][17].

Scalability, Rotation invariance and Affinity are three dominant aspects that every feature extractor should acquire. Through camera viewpoint, the traffic signs don't seem to be steady, therefore, it is mandatory for these feature extractors to be scale, rotation and affine invariant. The behaviour of various feature extractor methods with respect to certain parameters is displayed in figure 1[15][16][17].

TABLE I
DISTINCT FEATURE DETECTOR ALGORITHMS WITH THEIR SPECIFICATIONS

Feature Detector Methods	Year	Corner	Scale	Affine
Moravec	1979	*		
Harris and Stephens	1988	*		
Susan	1995	*	*	
Harris affine	2002	*	*	*
SURF	2006		*	
AGAST	2010	*	*	
BRISK	2011	*	*	
KAZE	2012	*		
TILDE	2014	*	*	

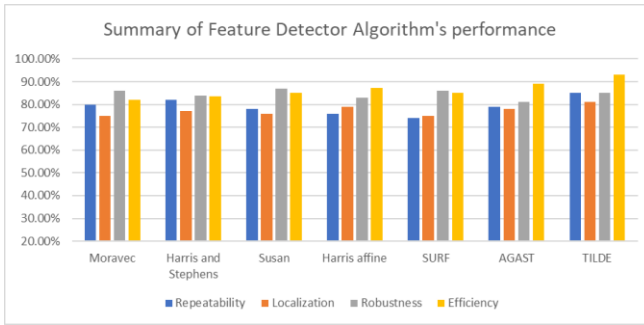


Fig. 1. Performance Summary of Feature Detector Algorithms

Analyzing from figure 1, KAZE and TILDE proves to be more powerful and proficient for object localization. While experimenting the above developed technologies for real time traffic sign detection and recognition, they showed considerable decline in the efficiency. Studies proved that the decline occurred due to the severe alterations in the contrast, lightening and rotation angles in real time images/videos of traffic signs. To attain improved performance, the hand-crafted feature extractor algorithms were combined with various learning algorithms like Support Vector Machine (SVM), Principle Component Analysis (PCA), Naïve Bayes, Gradient Boosting Algorithms and Decision Tree. Before the advancements in CNNs, (SVM) along with HOG, Deformable Part Models (DPM) and other Hand-crafted features were the highest development in object detection [18].

With the expansion in conventional Machine Learning based CNN algorithms, the feature extraction method has been

evacuated. The new Deep Learning approach detects the objects fundamentally through multiple layer beside containing the feature extraction layer.

III. OVERVIEW OF SSD, FASTER RCNN AND YOLOV2

Conventionally, Object Detection algorithms use one section of their network devoted to generate region proposals and then localize the objects in a image. These algorithms don't focus on the whole image, rather, the segment of image which have great likeliness of including the object. In 2016, Ross Girshick designed Faster RCNN, which is one such object detection algorithm. Faster RCNN is the latest advancement of RCNN family and uses Region Proposal Network (RPN) with CNN to generate region proposals instead of selective search algorithm used by RCNN and Fast RCNN [19]. These improvement makes Faster RCNN further accurate and speedy.

Recent advances in object detection includes algorithms like SSD and YOLOv2 which are much diverse from region proposal-based algorithms. In SSD, a singular convolutional neural network anticipates the bounding boxes and classifies the object into their respective classes. The basic variation amidst YOLOv2 and Faster RCNN is as described: (1) a single neural network does the feature extraction, training, detection and categorization in YOLOv2, it's an isolated end-to-end architecture; (2) YOLOv2 relate target detection as a regression count [20]. The target detection, object classification and confidence probabilities can be attained as soon as the image is given as an input to the model using a single neural network. The object detection action of the YOLOv2 model is explained in section A and section B.

A. Bounding Box Generation

Firstly, the input image is divided into $N \times N$ grid. The traffic sign is detected, if it is present in the particular grid. Individual grid detects K bounding boxes containing the traffic sign and their respective confidence probabilities. Total number of detected bounding boxes for each image are $N \times N \times K$ [21][22].

Individual bounding box anticipates 5 elements (X , Y , W , H) and a confidence value. The confidence value depicts the probability of box containing the object and its accuracy, W and H resembles the width and height of the bounding box and X , Y represents the offset of the midpoint of the bounding box corresponding to the grid.

B. Conditional Class Probability

Individual grid anticipates the conditional class probability to predict the likeliness of a particular traffic sign to a particular class, provided that the particular grid consists of traffic sign. The class confidence score is calculated for each bounding box by multiplying conditional class probability with the predictive values of distinct bounding boxes [23]. The above calculated Traffic Sign Score describes the probability of traffic sign detection in the bounding box and also the similarity between the box and the detection goal. The equation for confidence score is shown in (1).

$$\text{Confidence Score}(\text{obj.}) = \text{Prob}(\text{Obj.}) * \text{IOU}_{\text{Pred}}^{\text{Truth}} \quad (1)$$

In the above equation, $\text{Prob}(\text{Obj.})$ stands for the probability of the target object collapsing in the bounding box. shows the overlying estimate of the achieved bounding box to the ground truth box.

$$\text{IOU}_{\text{Pred}}^{\text{Truth}} = \frac{\cap(\text{predicted box}, \text{ground truth box})}{\cup(\text{predicted box}, \text{ground truth box})} \quad (2)$$

If the $\text{Confidence Score}(\text{Obj.})$ is zero, it depicts that the grid does not contain any targeted object.

IV. EXPERIMENTAL DETAILS.

A. Architecture and Setup Used

In this work, the evaluation and comparison of three architectures namely YOLOv2, Faster RCNN and SSD is recorded on the recently designed standard parameters: mAp (mean Average Precision) and FPS (Frames Per Second). The implementation of these architectures is performed on TensorFlow Environment with Anaconda 5.1.0 windows platform and the conclusions are appraised on Tensorboard. To perform comparison on identical environment, the YOLOv2 darknet model is converted to TensorFlow by modifying Darknet weights (.weights) and Darknet model(.cfg) to TensorFlow weights(.ckpt) and TensorFlow graph(.pb,.meta) respectively.

SSD, Faster RCNN and YOLOv2 are combined with various pre-trained CNN models like Pets, Coco, Inception to enhance their performances. In this paper, Inception V2, Resnet and Coco CNN models which are trained on ImageNet Database are used with SSD, Faster RCNN and YOLOv2 respectively.

B. Training Dataset

The Dataset is gained from German Traffic Sign dataset [24] which is most conventionally used Traffic Sign Dataset. Some sample images from the dataset are shown in figure 2. The dataset encloses different traffic signs with distortions like sunlight reflections, wind effects, barricades and occlusions. The dataset acquired consists of 20000 images for training and 1836 images for testing purpose. The images are classified into 5 groups: Pedestrian crossing, No entry, Speed limit, Stop sign and Turn ahead. SSD, Faster RCNN and YOLOv2 are trained and evaluated on the aforementioned dataset. In this paper, the above-mentioned three architectures are trained with 7832 images and 530 images are used for verification objective.

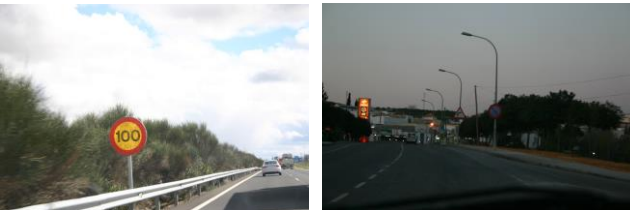


Fig. 2. Samples from the Traffic Sign Dataset

C. Experimental Results

In figure 3, graphs (a), (b) and (c) describes the loss factor of SSD, Faster RCNN and YOLOv2 respectively. The x-axis depicts the number of iteration loops and y-axis represents the loss factor. As depicted from the graphs, SSD lags in accuracy as well as speed when compared with Faster RCNN and YOLOv2. The loss factor of SSD at 3432 iteration steps is 0.95. Whereas, the loss factor of Faster RCNN and YOLOv2 is 0.43 and 0.33 at iteration step 5562 and 3497 respectively. The above figure shows that YOLOv2 is 68% faster than Faster RCNN and provides identical accuracy with lesser iteration loops.

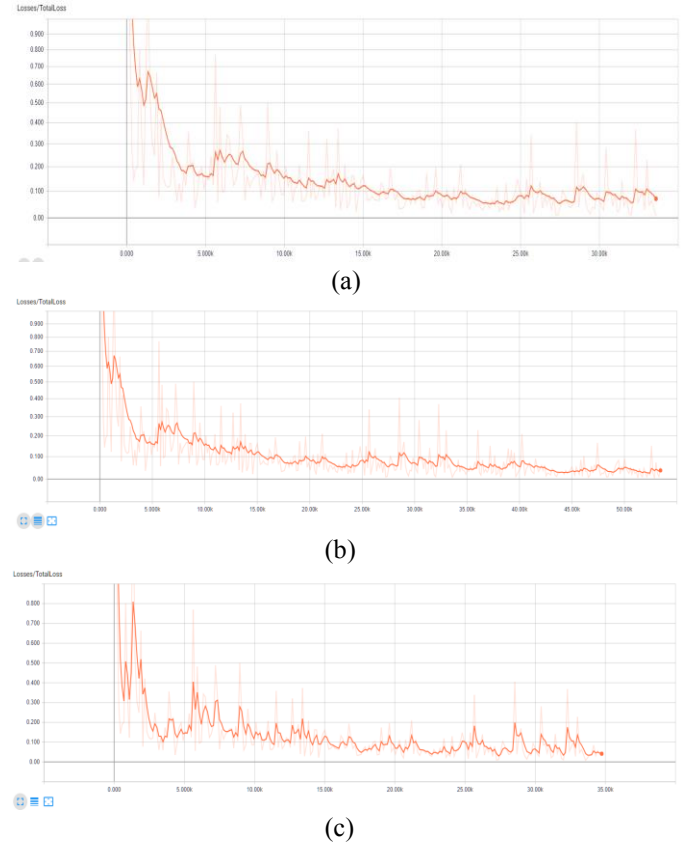


Fig. 3. Loss Factor of Architectures in CNN (a)SSD (b) Faster RCNN and (c) YOLOv2

D. Evaluation Parameters

The parameters on which the aforementioned three architectures are evaluated include mean Average Precision(mAp) and Frames Per Second(FPS). Mean Average Precision is a prominent metric to evaluate the degree of accuracy of object detectors and classifiers. It is average of

three sub-parameters: Precision, Recall and IOU. Precision describes the exactness of the predictions i.e. the ratio of true predictions. Recall evaluates how acceptable are entire positive predictions. The mathematical formula for Precision and Recall are given in equation (3) and (4).

$$Precision = \frac{True\ Positive(TP)}{True\ Positive(TP) + False\ Negative(FN)} \quad (3)$$

$$Recall = \frac{True\ Positive(TP)}{True\ Positive(TP) + False\ Positive(FP)} \quad (4)$$

In the above equations, TP, FP and FN depicts the total count of predictions that perfectly identified the traffic signs, total count of predictions that recognized other objects as traffic signs and total count of predictions that identified traffic signs as other objects. The True Positive, False Positive and False Negative values for all 150 testing images of all three architectures is represented in table 2. The Precision, Recall and IoU of all three architectures are represented in table 3.

IoU(Intersection over Union) represents the overlap percentage of the predicted bounding box to the ground truth box. In general, an IoU threshold (commonly 0.5) is defined which concludes the prediction as True Positive and False Positive.

TABLE 2
TRUE POSITIVE, FALSE POSITIVE AND FALSE NEGATIVE IMAGES
FOR ALL THREE ARCHITECTURES

Architectures	Total Images	TP	FP	FN
SSD	150	110	24	16
Faster RCNN	150	130	22	8
YOLOv2	150	140	18	3

TABLE 3
COMPARISON OF PRECISION, RECALL AND IOU OF ALL THREE
ARCHITECTURES

	SSD	Faster RCNN	YOLO
Precision	87.30%	94%	97.90%
Recall	82.09%	85.53%	88.61%
IOU	68.40%	70.40%	75.70%
mAp	68.35%	74.68%	77.89%

FPS(Frames Per Second) describes the number of frames grasped per second at the training time. It depicts the speed of a particular algorithm. The performance parameters and their comparison of all three architectures are shown in figure 4.

E. Interpretation of Results

SSD, Faster RCNN and YOLOv2 are compared and evaluated with the support of their mAp(Precision, Recall and IoU) and FPS. The average mAp of all five classes of SSD, Faster RCNN and YOLOv2 are 68.35, 74.68 and 77.89 respectively. On overall comparison, SSD is 21% less accurate

than Faster RCNN and YOLOv2. The mAp result of Faster RCNN and YOLOv2 is collateral, YOLOv2 outperforms Faster RCNN by 3.5%. When FPS is considered, Faster RCNN lags from both SSD and YOLOv2. Since, SSD and YOLOv2 are regression architectures and work with a single neural network, they are 4 times faster than Faster RCNN. Analysing YOLOv2 and SSD in terms of FPS, YOLOv2 is 16% faster than SSD. Overall comparison of all three architectures proves that YOLOv2 is faster and accurate than both Faster RCNN and SSD. The sample of the output are shown in Figure 5.

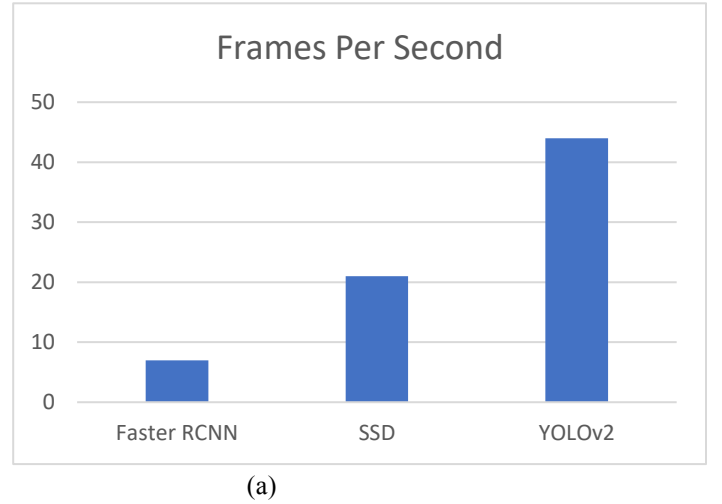
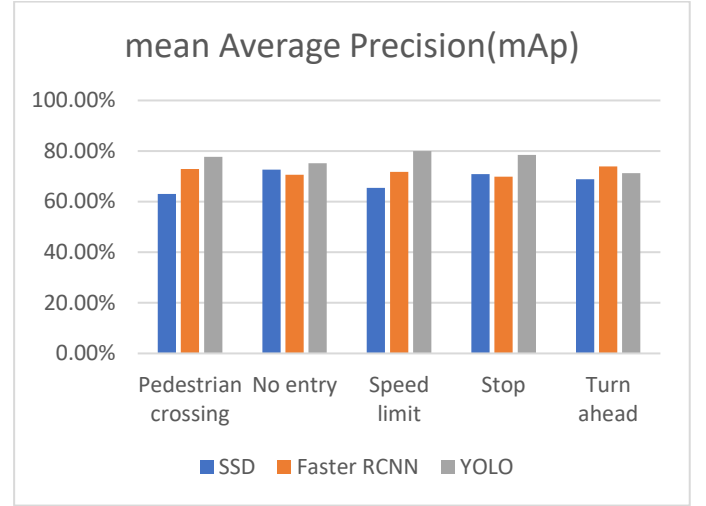


Fig. 4. Comparison of Faster RCNN, SSD and YOLOv2 with respect to parameters (a) mAp and (b) Frames per second





Fig. 5. Samples from the Result

V. CONCLUSION

In this work, SSD, Faster RCNN and YOLOv2 are examined with three distinct pretrained CNN models. SSD, Faster RCNN and YOLOv2 were assembled with Inception v2, Resnet and Coco CNN model respectively. The implementation of all three architectures was performed on TensorFlow framework with Anaconda 5.1.0 windows platform and the conclusions were appraised on Tensorboard. The accuracy and the speed of these architectures are evaluated using conventional and standard parameters: mAp(Precision, Recall and IoU) and FPS. On comparison, SSD lags behind Faster RCNN and YOLOv2 in terms of both accuracy and speed. While, YOLOv2 comes out to be 3.5% more accurate than Faster RCNN. Also, the learning rate of YOLOv2 is 68% faster than Faster RCNN and 16% speedy than SSD. But, when it comes to detection of very small traffic signs, YOLOv2 will not perform upto the mark and Faster RCNN will outperform YOLOv2 in terms of accuracy. So, the performance of YOLOv2 will get limited after a certain extent of object size.

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